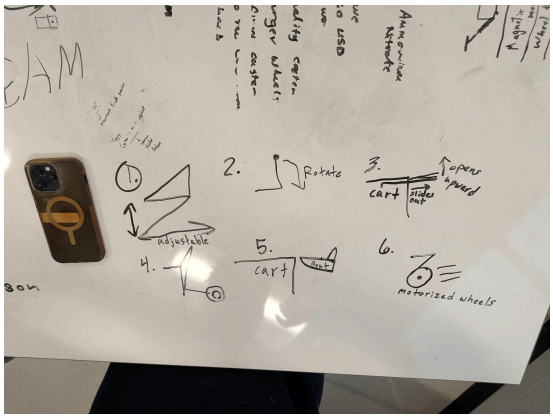
Brijodison

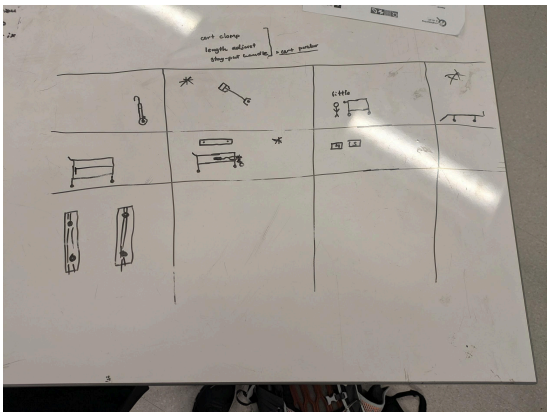
Engineering Notebook Element D

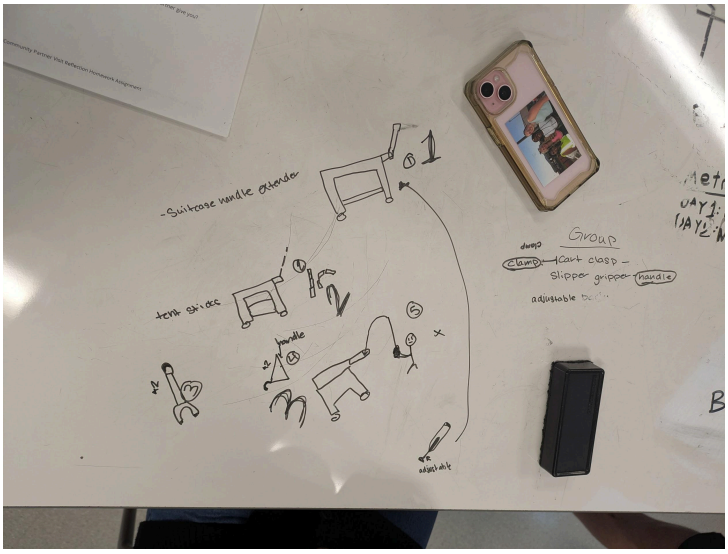
Engineering Service Learning

Design Concept Generation

At the start of our prototyping Mr. Kiser told us to come up with at least four designs each that we thought would answer the prompt the best. He then wanted us to share our ideas with our groups and decided which ones would work the best. After all three of us shared, we picked out the ideas that we thought would work the best in this situation. You can see a few examples of the designs we ideated below.







Some of the ideas we came up with at our table were making a suitcase handle, a triangle handle, a hinged handle etc. as you can see in the pictures above. We brainstormed a few other designs that were not entirely feasible (e.g. boat) in hopes of inspiring further development although we eliminated these designs very rapidly as they did not meet the criteria or constraints. (In terms of the boat design, it was fairly difficult to source a motor boat for $30 that would operate on land.) The next day, we then had to create a design matrix to figure out which design we thought would work the best. While making the matrix we decided to base it off of many different items like: time to assemble, size, force needed, height range, and angle bent over at.

We adapted many of the most important parts of the criteria that could be easily determined without deep analysis of the issue. For instance, we did not include the stability because it would require fairly in depth calculations for each design to determine the force needed to budge it. We determined this would be better understood experimentally, making modifications to our design later to resolve issues. We consequently only included criteria that were inherently linked to the form of the design. Assuming no major revisions, we will always assemble the design the same way, so time to assemble should not be able to drastically change. Likewise, the size cannot drastically change once we have created the prototype. Without changing the angle at which the force is applied, requiring a modification to the overall form of the design, the force needed would not be able to change. Similar ideas may be applied to understand the presence of height range and angle bent over. On the other hand, properties like the stability could be modified without significant revision to the product (e.g. by adding rubber to the product as we did), so we did not include it on the decision matrix.

Time to assemble	0.1	How long the product takes to attach to the cart (0 denotes 5 minutes; 10 denotes 10 seconds)

Size	0.1	The smallest box the product could fit inside when under normal usage (i.e. NOT exact volume) (0 means it's around 1/2 the size of the cart; 10 means it's around 1/10)
Force needed	0.25	The force needed to push the cart (0 means it makes the cart maximally harder to push; 5 means it makes no difference; 10 means it makes the cart maximally easier)
Height Range	0.3	The range of heights it can adapt to (0 means it can't adjust; 8 means it handles anybody taller than the device perfectly and 10 means it handles anybody over 4'10) (https://www2.census.gov/library/publications/2010/compendia/statab/130ed/tables/11s0205.pdf)
Angle bent over	0.25	Angle Mr. Kiser would have to bend over to assemble apparatus (0 is much greater than 90 & 10 is within normal standing range)
sum (should = 1)	1	

After summating the responses from three of our group members, we came to the following results. Skis emerged as a clear winner with one particular problem—it did not fit entirely within the constraints. We decided it would be too difficult to model the skis solution for

a cheap price without access to different machinery to facilitate the development process, so we moved to the second highest design, the suitcase handle.

	Sliding Handle	Suitcase Handle	Triangle Handle	Hinged Handle	Skis	Grippy Toy
Raw Total	16.8	16.95	16	16.4	17.7	14.7
Total (out of 10)	5.6	5.65	5.333333333	5.466666667	5.9	4.9

At the start of our prototyping we tried many different ways that we could make a suitcase handle cart extender. We started off trying to brainstorm different materials that we could use to make it and the most effective way. After looking around the classroom and trying to use many different items it never seemed to go the way we tried and never seemed to work. After we had a group discussion about our idea, we cancelled it and began to come up with new ideas that would better fit the requirements needed.

While searching in our design matrix for another idea to attempt we realized that none of our solutions really fit the requirements that we wanted to reach. We quickly “took it back to the drawing board” and started brainstorming again. Through the many attempts and ideas we came up with one that stuck.

We came up with an idea of a product that looks like two ski poles that can attach to the cart, but has a longer handle. After running it through our design matrix we realized that it better fit all the requirements. We saw how it wouldn’t take too long to assemble, it could reach multiple heights, it wouldn’t take too much force, and it kept Mr. Kiser’s back up right to limit stress. It also seemed to solve other problems like having to make it adjustable. This solution clearly matched our criteria far better than any other. Adding the solution as a column to the

decision matrix made it a clear winner as shown below. Using the final score out of ten from the decision matrix as a proxy for meeting our criteria, the two pole solution fits very well. Once we all were satisfied with the idea we began to start creating.

	Sliding Handle	Suitcase Handle	Triangle Handle	Hinged Handle	Skis	Grippy Toy	Two Pole
Raw Total	16.8	16.95	16	16.4	17.7	14.7	24.45
Total (out of 10)	5.6	5.65	5.333333333	5.466666667	5.9	4.9	8.15

While we were bringing this idea to life, we realized that it was more effective and overall just a better and more successful product. Throughout the rest of our designing process we kept checking with each other and our design matrix to make sure that it was all going the way we planned and the way we wanted it to. After a while of creating we took a break and talked to each other to make sure that everybody was on the same page and still wanted to continue with this design. After hearing feedback from our group and others we were finally ready to continue. When we were assured that everybody liked the way our design was going we continued to create and check in.

We created a general plan of action to continue with the prototype. We would determine the sizing of the pipe based on the dimensions of the handle, and we would determine the sizing of the clamp based on the dimensions of the lip, but we would determine the rest of the sizing by the availability of materials and fit with the cart. Instead of measuring the angle at which we needed to drill a hole in the support to fit it properly, for example, we would instead directly reference the necessary angle to ensure the prototype fit the situation. We figured it would only

be worth the time to CAD the product once we had determined whether or not it was viable by prototyping a proxy solution first. We thus decided to start with the base and continue adding parts per our design until the prototype matched the design. Upon this step, we would go through a series of tests and revisions to find a final prototype that would inspire the design of our true solution.

Once we finished creating our prototype we went back through our design matrix to make sure that we accomplished what we wanted and that it followed all of our requirements. We also took feedback from others like Mr. Kiser and the engineers that came in on our “designathon”. Once we finished we felt that our current design was our best solution and answer to our prompt and design matrix.